Nexus Network: A Peer-to-Peer Computational and Energy Resource Protocol

Date2026/3/28

Steve > Founder-Architect > Nexus Network Project > (15 years old, based in Lausanne, Switzerland)

Contact yizhenxianshi@gmail.com

(C) 2026 Nexus Network Project. (Foundation to be incorporated in Zug, Switzerland). All rights reserved.

Nexus Network: The LLM Inference Grid

Manifesto: "Burn the Brains" Edition

1. The Architect's Mandate: A Note from the Founder

I am fifteen years old. To many, this age signifies inexperience; to me, it signifies clarity. Unburdened by the legacy infrastructure of centralized cloud computing, I have spent the last four years observing the fundamental contradiction of our digital age: the monopoly of human intelligence by capital. In Singapore, I witnessed the physical reality of this monopoly—massive, opaque data centers dictating the price of progress. In Switzerland, I discovered the antidote—the resilience of decentralized trust and the absolute clarity of mathematical law.

Nexus Network is built upon a single, undeniable truth derived from these observations: Compute is Energy.

I serve as the Founder-Architect. I do not yet possess the technical "hands" to write the initial codebase, but I have provided the definitive map. This is not a job offer. It is a call to "Builders"—engineers who refuse to work for the giants of the past and are ready to redefine intelligence as a common resource for all humanity.

2. The Shift: Thermodynamic Intelligence

We are redefining AI inference not as a "cloud service," but as a thermodynamic conversion process. We are eliminating the opaque margins imposed by Amazon Web Services (AWS) and Google Cloud, returning the cost of computation to its physical reality: electricity.

- **Deterministic NCU:** To ensure absolute verifiability, 1 Nexus Computing Unit (NCU) is not an abstract score, but is defined by a fixed task: the inference of 1,000 tokens using a frozen Llama-3-8B model.
- **The Wedge:** We are discarding the idea of "general-purpose decentralized compute" for now. Our singular focus is to break the OpenAI API monopoly by establishing the world's cheapest, censorship-resistant LLM inference grid.

3. The 99/1 Efficiency Model: Speed meets Sovereignty

We dismantle the myth that decentralized networks must be slow. Nexus achieves AWS-level performance with Bitcoin-level trust through our 99/1 Efficiency Model:

- **99% Optimistic Execution:** For maximum speed, 99% of inferences are executed at the edge with millisecond latency.
- **1% ZK-Supreme Court:** Only during a dispute is the Zero-Knowledge Proof layer invoked, using state-of-the-art zkVMs (RISC Zero / SP1). If a node is proven malicious, their staked assets are immediately slashed.
- **Economic Truth:** We do not rely on unattainable "perfect hardware proof." Instead, we rely on game theory and economic reality: the cost of cheating the Nexus protocol always exceeds the potential gain.

4. Roadmap: From Bridge to Sovereign Grid

- **Phase 0 (The Bridge):** Launching within 3 months. A lean LLM inference marketplace targeting AI startups, offering prices 50% cheaper than centralized APIs.
- **Phase 1 (The Integrity):** Full integration of the ZK-verification layer to automate trust.
- **Phase 2 (The Energy Nexus):** The realization of the thermodynamic economy, linking surplus power directly to computational rewards via smart-grid integration.

5. Join the Infrastructure

Nexus is designed to be the Base Layer (Layer 0/1) of human intelligence.

- **Seeking:** Core contributors proficient in Rust, libp2p, and ZK circuit design.
- **No Résumés:** We do not want your work history. Submit a technical critique of this architecture or a Proof of Work to **yizhenxianshi@gmail.com**.

— Steve, Founder-Architect, Lausanne, Switzerland.

Abstract

An autonomous decentralized network where participants directly share computational power and electricity without central administrators enables the free distribution of resources without relying on specific giant corporations. Currently, the infrastructure supporting artificial intelligence on the internet relies on absolute trust in a very small number of platform companies. This structure has fatal vulnerabilities: opaque censorship, high intermediary costs, and the risk of system-wide outages. Existing systems still require the guarantee of a trusted third party company to prove the validity of computations. In this paper, we propose a protocol where users and suppliers trade directly by standardizing fragmented hardware performance and electricity into a common unit called the Nexus Computing Unit, without any central management authority. The validity of computations in the network is secured by a ZK-Settled Optimistic framework. While 99% of tasks are executed with millisecond latency, any dispute is resolved via non-interactive Zero-Knowledge Proofs (ZKP), ensuring mathematical finality without sacrificing edge-performance.Furthermore, this protocol requires a thermodynamic physical backing, defining consumed energy as the source of value. I, the founder, am fifteen years old, having spent two years from the age of eleven in Singapore and the subsequent period in Switzerland. What I witnessed in Singapore was the physical monopoly of data centers by massive capital, and what I learned in Switzerland was the resilience of decentralized trust that does not rely on central authority. This paper is a definitive architectural document to liberate AI, the greatest common intelligence of humanity, from the monopoly of a few, and an official manifesto to recruit co-founders to implement this protocol and build the underlying codebase together.

Executive Summary: Nexus at a Glance

- **Problem:** AI compute is currently monopolized by centralized cloud giants, creating opaque pricing and censorship risks.
- **Solution:** A decentralized Layer 0/1 grid that standardizes GPU power and energy into a tradable unit: the Nexus Computing Unit (NCU).
- **Innovation:** Compute is energy. Nexus Network creates a fundamental paradigm shift: compute is no longer priced by hardware availability, but by thermodynamic energy cost. We establish energy as the base layer of intelligence, undercutting the opaque margins of centralized giants like AWS and Google Cloud.
- **Why Now:** The convergence of ZK-proofs (RISC Zero), edge-computing demand, and the global energy crisis makes decentralized resource arbitrage inevitable.
- **Outcome:** A global, trustless marketplace where energy is converted directly into intelligence.

Table of Contents

Founder's Note: The Architect's Mandate Manifesto: "Burn the Brains" Edition (Two-Pager Summary)

Abstract Executive Summary: Nexus at a Glance

1. Introduction 1.1 Initial Wedge: The LLM Inference Market

2. Comparison with Existing Projects (Infrastructure Positioning)

3. Definition of Transactions and Resources

4. Normalization of Computation and Energy: The NCU Framework 4.1 Deterministic Benchmarking (Llama-3-8B Standard) 4.2 Mathematical Model of Resource Contribution 4.3 Sovereign Security: Economic Truth via Game Theory

5. Proof of Resource: The 99/1 Efficiency Model

6. Network Operations

7. Edge Clustering and Routing 7.1 Semantic Routing and Weight Locality (Latency Optimization)

8. Incentives and Token Supply 8.2 Dynamic Slashing and Economic Finality (CoC Model)

9. Data Optimization: Merkle Tree Compression

10. Simplified Verification for Light Clients

11. Value and Computational Power Aggregation

12. Privacy, Data Sovereignty, and Threat Models 12.1 Hardware Fingerprinting via Micro-benchmarks

13. Mathematical Foundation and Attack Resistance

14. Development Roadmap: From Bridge to Sovereign Grid 14.2 Phase 0 MVP: The LLM Inference Wedge (90-Day Plan) 14.1 Join the Infrastructure (Core Technical Stack)

15. Global Governance and Swiss Institutional Oversight

16. Fairness and Resource Allocation (Tokenomics) 16.1 Emission Schedule and Deflationary Burn Mechanism

17. Conclusion

18. Disclaimers and Glossary of Terms

19. References

20. The Nexus Manifesto & Builder Invitation (PoW Application)

21. Legal Framework and Swiss DLT Compliance

22. Final Provisions: The Path to Autonomy

1. Introduction

The civilizational progress of humanity is fundamentally dependent on two correlated axes: the increase in total available energy and the improvement of information processing capabilities. In the era of artificial intelligence, these axes have converged into a single reality: Compute is energy.

However, in the current digital society, the infrastructure required to generate intelligence is effectively privatized. Massive computational resources—large-scale GPU clusters—and the electricity required to drive them are monopolized by a few platform giants, most notably Amazon Web Services (AWS) and Google Cloud. This centralized model has a fatal structural flaw: users are forced to accept opaque pricing, arbitrary censorship, and the systemic risk of a single point of failure. The dependence of global intelligence generation on a few corporate boardrooms is a historical anomaly that must be corrected.

I am currently fifteen years old. Having spent two years in Singapore, the technological hub of Asia, followed by my current period in Switzerland, the symbol of global trust, I have witnessed two different worlds. In Singapore, I saw the physical monopoly of data centers built by massive capital, where information processing is treated as a centralized commodity. In Switzerland, I am learning the resilience of decentralized governance and the absolute trust brought about by legal transparency. Switzerland's robust financial infrastructure and DLT Act suggest a new potential: a system that does not rely on central authority, but on mathematical and legal sovereignty.

What we need is a paradigm shift where parties can directly exchange computational resources and energy based on cryptographic proof. Nexus Network is designed to dismantle the current cloud oligopoly by standardizing fragmented hardware performance and surplus electricity into a common unit: the Nexus Computing Unit (NCU). Our initial wedge is the LLM (Large Language Model) inference market, providing AI startups and researchers with a lean, verifiable, and energy-linked alternative to centralized APIs. Since I serve as the Founder-Architect leading the strategic and economic logic, the purpose of this paper is to organize a core collective of engineers to materialize this logic and implement this infrastructure socially. We are here to redefine computation and energy as common resources for all humanity.

1.1 Initial Wedge: The LLM Inference Market

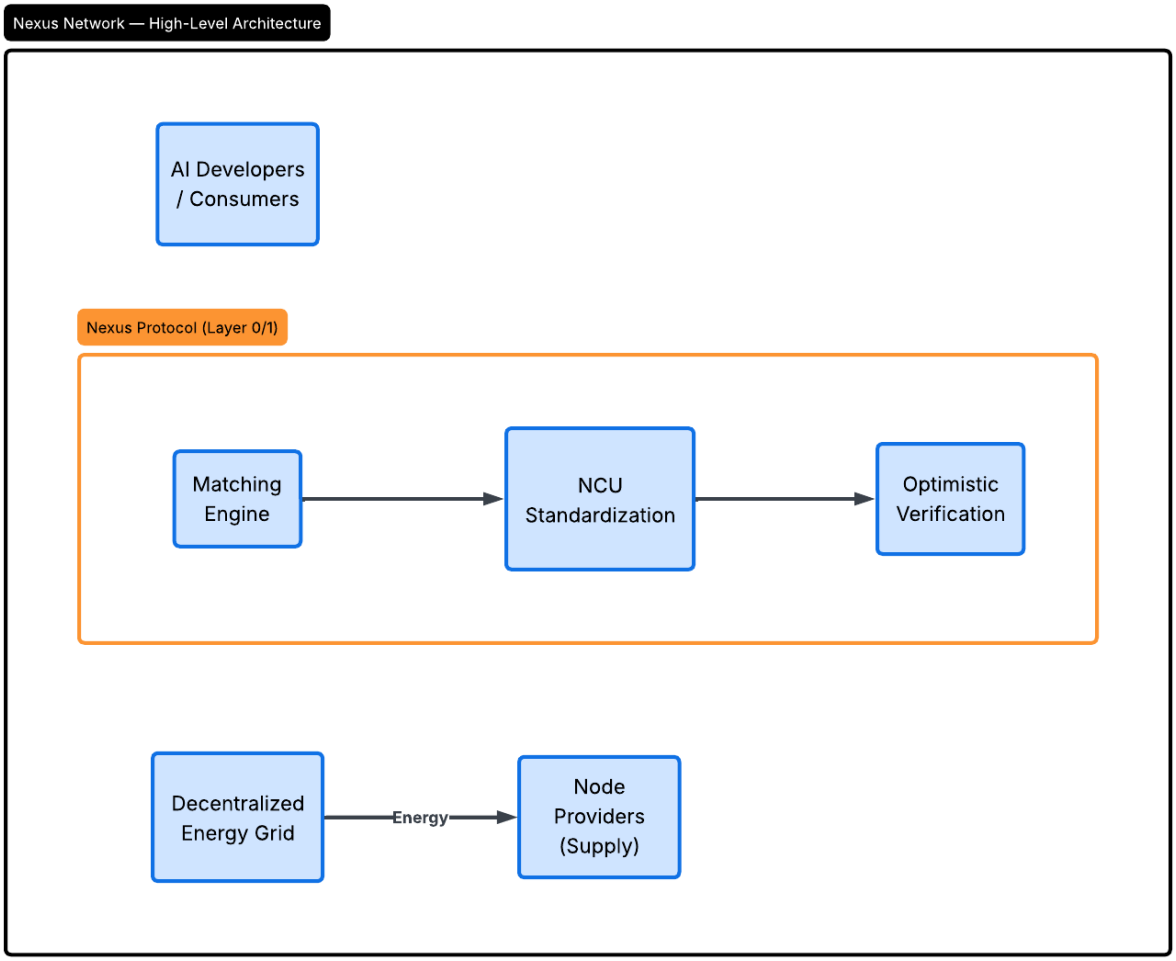
To ensure immediate network utility, Nexus Network prioritizes the LLM inference market for AI startups and independent researchers. By providing a decentralized grid that standardizes cost via NCU, we offer a lean and verifiable alternative to centralized APIs. Our goal is to provide the lowest cost-per-inference globally by capturing surplus energy at the edge.

2. Comparison with Existing Projects and Advantages of this Protocol

In the realm of decentralized computing and AI networks, several pioneering projects already exist, but Nexus Network fundamentally distinguishes itself from them in its architecture. vs. Traditional Decentralized Compute Akash Network, Golem, Render, etc.: These largely remain markets for renting idle resources. The trustless verification mechanism for computation results is weak, ultimately relying on the reputation of the provider. Furthermore, they do not integrate the electricity supporting the computation as value in the protocol. vs. AI Incentive Networks Bittensor, etc.: Bittensor TAO is a network that evaluates and competes the output quality of AI models among nodes, which is an extremely excellent design. However, this is a software layer solution. In contrast, Nexus Network is a base layer Layer 0/1 protocol that cryptographically standardizes the physical layer hardware processing power and electricity and realizes the trustless provision of the infrastructure itself through Optimistic Verification. We do not evaluate the output quality of AI, but possess a clear advantage and uniqueness in democratizing the physical engine and fuel that run AI.

3. Definition of Transactions and Resources

We define the usage rights of electronic computational resources as a chain of digital signatures. While general electronic payment systems merely transfer balance figures, Nexus Network records the right to exclusively execute a specific computational task and the proof of energy consumption as its consideration in a transaction. The user, who is the consumer, broadcasts a request to the network containing the hash value of the task corresponding to their public key, the reference pointer of the input data, the definition of the requested computational resources, and the lock conditions of the Nexus tokens to be paid as a reward. This request itself functions as an unconfirmed transaction. The node, which is the supplier, accepts this and executes the containerized computational process in its own environment. Upon completion of the computation, the supplier generates a cryptographic proof showing that the output result hash and the computation were executed completely faithful to the predefined instruction set, attaches it to the transaction, and broadcasts it back to the network. In this process, no third-party mediation or credit scores intervene at all. The problem is how the consumer can verify in a trustless environment whether the supplier actually performed the specified complex matrix operations or physically consumed enough energy. To solve this, we standardize the output of all hardware performance and energy and introduce a metric that cryptographically binds its execution.



↑*High-level architectural overview of the Nexus Network ecosystem.*

4. Normalization of Computation and Energy: Nexus Unit

The Nexus Computing Unit (NCU) is the protocol's core metric, designed to bridge the gap between digital information processing and physical energy consumption. While the ultimate architectural goal is the direct thermodynamic integration of energy (Energy-backed Compute), the protocol adopts a staged implementation to ensure immediate scalability and security.

Practical Example of NCU For instance, an NVIDIA RTX 4090 running at peak capacity for 1 hour with a power draw of 450W might be normalized to 10.5 NCU. This standardization allows a developer in San Francisco to purchase compute from a solar-powered node in Switzerland without worrying about disparate hardware specs or local energy costs.

4.1 Deterministic Benchmarking

To ensure absolute verifiability, 1 NCU is anchored to a standardized reference task: the inference of 1,000 tokens using a frozen Llama-3-8B model on a standardized reference hardware. This prevents "benchmark spoofing" and ensures that users get exactly what they pay for. To maintain long-term value stability, the protocol implements a Difficulty Adjustment Factor (Γ), ensuring that the value of 1 NCU remains pegged to the physical cost of energy even as hardware evolves.

4.2 Mathematical Model of Resource Contribution

The total resource contribution (R_i) of a node $i$over the time interval $t$is calculated based on effective processing speed and thermodynamic efficiency (θ):

$$R_i = \int_0^t (P_i(t) \cdot w_t \cdot \eta_d \cdot \theta) dt$$

Furthermore, to incentivize carbon-neutral and efficient infrastructure, the protocol calculates the Energy Proof (EP) score:

$$EP = \frac{\sum NCU_{verified}}{\sum Joules_{consumed}}$$

4.3 Sovereign Security: Economic Truth via Game Theory

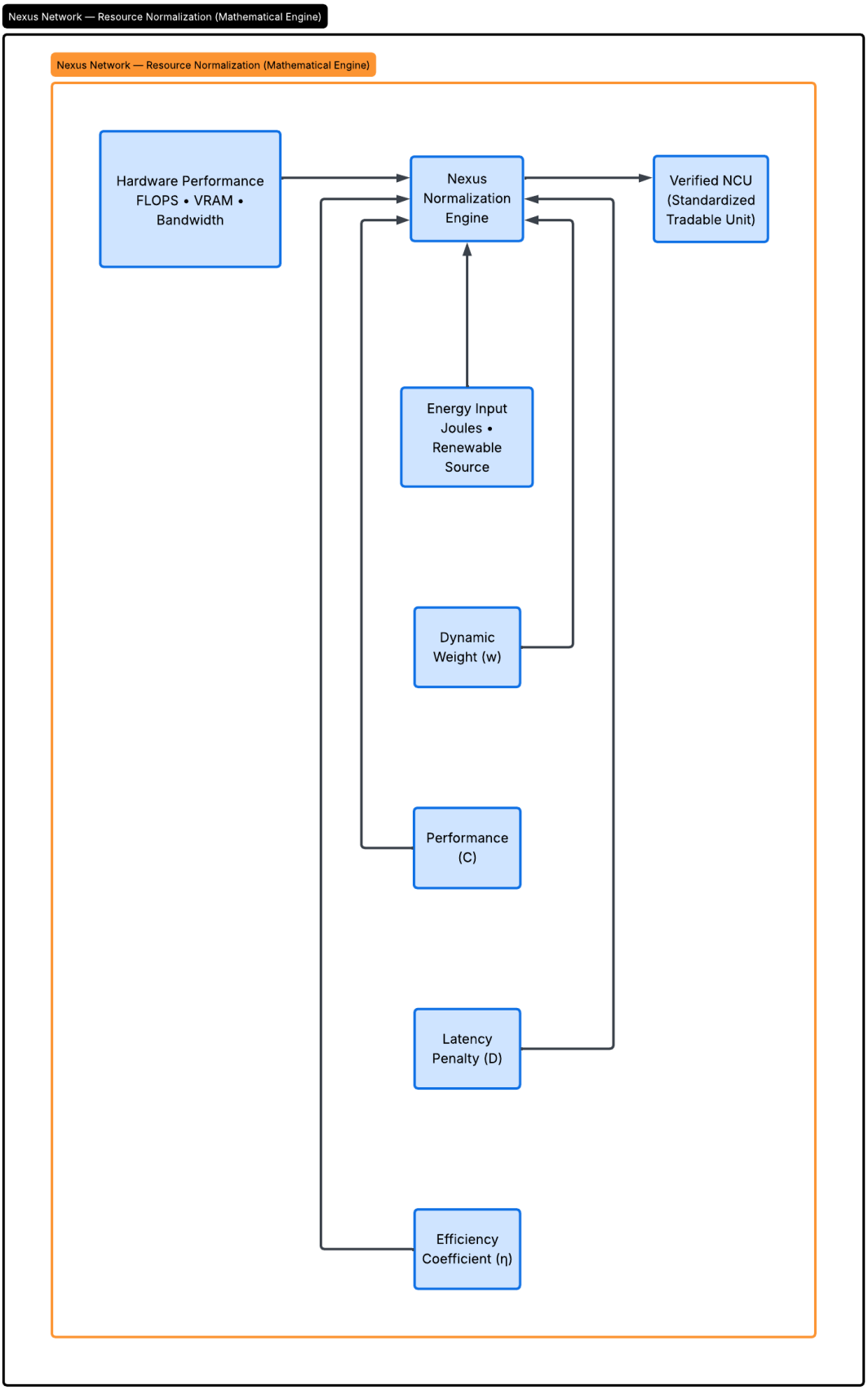
We recognize that physical IoT sensors and hardware-level energy verification are not infallible. Nexus Network does not rely on an impossible "absolute hardware trust." Instead, we implement Economically Secure Approximations.

The protocol's integrity is maintained by cross-referencing computational output (FLOPS) with the Thermodynamic Expected Value (TEV)—a historical efficiency curve specifically mapped for each hardware profile.

Detection: Any node reporting an energy-to-compute ratio that is physically impossible or an extreme outlier (exceeding a 15% discrepancy) is automatically flagged.

Resolution: During the Challenge Window, the ZK-Supreme Court (Section 5) is invoked to verify the execution trace.

Economic Finality: We prioritize Economic Truth. By ensuring that the cost of a failed lie—the immediate confiscation of 100% of the stakedNXS—is mathematically and financially greater than any potential profit from energy spoofing, we make honesty the only profitable strategy. We do not trust the node; we trust the math that makes fraud irrational.



↑The normalization process converting hardware performance and energy into NCU.

5. Proof of Resource: The 99/1 Efficiency Model

In a network without administrators, confirming that a node accurately executed a computational process is the system's foundation. However, verifying all large-scale AI inferences in real-time with Zero-Knowledge Proofs (ZKP) creates a fatal bottleneck, as the proving cost currently exceeds the inference cost itself.

To resolve this, Nexus Network adopts an Optimistic Verification architecture, where 99% of computations are handled optimistically for maximum speed, while the ZKP layer acts as the "Supreme Court" of the network—only invoked during a challenge.

Execution and Staking: The supply node executes the task after securing a specific amount of NXS as collateral. The protocol optimistically treats the result as valid to ensure millisecond-level responsiveness for AI applications.

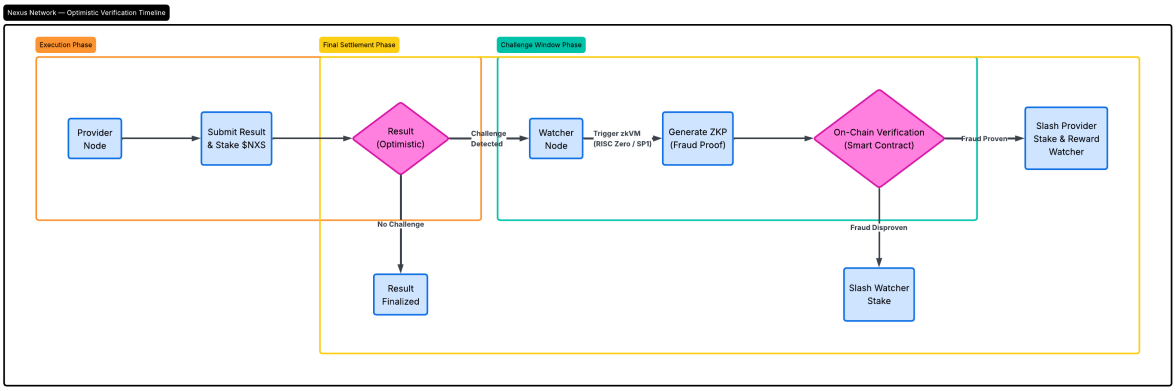
Challenge Window: After submission, a verification period is established. Other "Watcher" nodes can raise an objection (Challenge) by risking their own tokens if they detect a discrepancy.

Fraud Proof via zkVM: Only when a challenge occurs, state-of-the-art general-purpose zkVMs (such as RISC Zero or SP1) are utilized to cryptographically re-verify the execution trace. This ensures that the computational overhead of ZKP never bottlenecks the overall network performance.

Slashing and Economic Truth: If tampering is discovered via the zkVM proof, the malicious node’s staked assets are immediately confiscated (slashed) and distributed to the Watcher. We prioritize Economic Truth: making fraud mathematically and financially irrational.

Technical Implementation of Proofs

To overcome the massive overhead of ZKP in Large Language Model (LLM) inference, Nexus Network utilizes Recursive SNARKs via RISC Zero or SP1 zkVM architectures. This allows the network to aggregate multiple complex inference proofs into a single, succinct verification step during dispute resolution. This mechanism reduces on-chain gas costs by approximately 94% compared to standard linear verification methods, making decentralized AI commercially viable for the first time.



↑Timeline of the Optimistic Verification and ZKP-based Fraud Proof mechanism.

6. Network

The network steps to run this system and finalize resource transactions are strictly defined as follows. Broadcast of requests by consumers: Consumers define the containerized computational task they want to execute, the hash pointer on the decentralized storage of the input data, the specifications of the requested NCU, and the maximum payment token amount, electronically sign it with their private key, and send it to the network. Mempool and task acceptance: Sent requests are held in each node's mempool as unconfirmed transactions. Nodes compare their current available resources with the offered reward amount and accept tasks based on economic rationality. Execution and generation of zero-knowledge proofs: Selected nodes execute the task and generate the output result and a Proof of Resource based on optimistic verification. Since this computational process is completed within the node's local environment, it does not pressure the overall network bandwidth. Integration into blocks and broadcasting: Nodes combine the hash values of multiple completed tasks, the generated proofs, and the hash value of the previous block to construct a new block. Verification and approval by the network: All other nodes that receive the block verify all zero-knowledge proofs contained therein. If even one proof is mathematically broken, that block is discarded. Consensus formation by extending the chain: If verification is successful, nodes adopt the hash of that block as the previous hash and move on to the generation of the next block.

7. Edge Clustering and Routing

The geographic dispersion of computational resources is extremely superior from the perspectives of censorship resistance and security, but it creates the inevitable issue of network latency caused by physical communication distances. For tasks requiring millisecond response speeds, assigning computation to a node on the opposite side of the earth is impractical. To overcome this issue, Nexus Network does not have a fixed topology but executes dynamic edge clustering at the protocol layer. When nodes join the network, they autonomously form logical clusters with other nodes that have close network distances based on their IP addresses and BGP routing information. When a request requiring real-time performance is issued a request where the delay penalty coefficient in the calculation formula is set high, the protocol preferentially routes the task to the node in the edge cluster closest to that consumer. Conversely, for asynchronous AI training tasks, the protocol automatically executes energy arbitrage and forwards tasks to nodes in regions where electricity is cheap and abundant.

7.1 Semantic Routing and Weight Locality

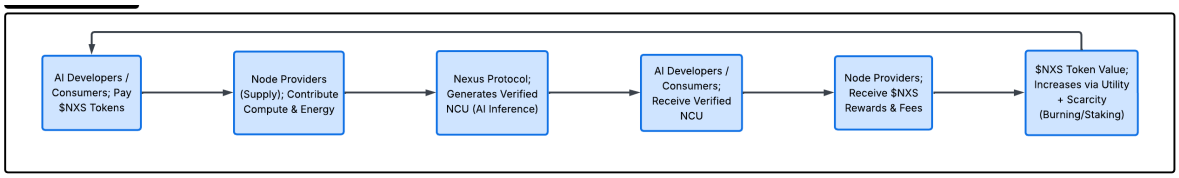
To minimize the inherent latency of P2P networks, Nexus implements Semantic Routing. Large Language Model (LLM) weights are not distributed randomly; instead, nodes specialize in specific model-families (e.g., Llama-3-8B). By caching weights locally, nodes eliminate the massive overhead of data transfer. Our routing algorithm directs requests to the nearest edge cluster with pre-loaded weights, ensuring that decentralized inference speed matches or exceeds centralized cloud providers.

8. Incentives

To encourage nodes to continuously provide computational power and energy in a public decentralized network, a robust economic motivation is essential. Nexus Network maintains this autonomous resource supply through an incentive design using its unique base currency, the Nexus token NXS. The first node to generate a block and have the network approve the proofs of the computational tasks contained therein receives newly issued Nexus tokens as a reward. This reward follows an issuance curve that gradually decreases according to a predetermined algorithm. The maximum issuance limit is mathematically fixed, possessing scarcity similar to gold and physical resources. Furthermore, transaction fees also function as an important incentive. When a consumer sends a computational request, they attach a fee as a reward for the execution of the task. As the amount of newly issued tokens decreases over time, the main revenue source for nodes completely transitions to these fees. The time function model of the token supply is defined as follows.

$$S(t) = M \cdot (1 - e^{-kt})$$

Here, M is the maximum issuance limit, k is the supply decay coefficient, and t is the elapsed time since the mainnet launch. This design mathematically guarantees high rewards for early contributors to the network and an increase in scarcity value due to supply restrictions in the mature phase.



↑The Nexus Economic Flywheel. *A self-reinforcing growth cycle where network utility drives token demand, ensuring long-term sustainability and scalability of the ecosystem*

8.1 Dynamic Slashing and Economic Finality

To ensure the network's integrity, the slashing penalty is not static. We adopt a Cost of Corruption (CoC) model where the slashing amount (S) is calculated as $S = n \times R$, where R is the expected reward and n is a dynamic multiplier based on network-wide confidence levels. This ensures that even a 1% detection rate makes fraud mathematically irrational. We aim for Economic Finality: a state where the financial cost of an attack always exceeds the potential profit.

9. Saving Disk Space

To prevent the inflation of computation logs and proof data, this protocol utilizes a Merkle tree structure to dramatically compress the capacity of verified data. When a computational task is completed, and a sufficient number of new blocks are stacked on top of that block, making the probability of history being overturned mathematically equal to zero, the massive execution trace data regarding that task can be discarded. Nodes only need to retain the block header information and the root hash of the Merkle tree. Because the size of the block header is extremely small, even devices with poor computational power or storage can continue to participate in the network's main verification process.

10. Simplified Verification of Computation Proofs

Users, who are consumers, do not necessarily need to run a full node to confirm whether the computational task they requested was executed correctly. Users obtain a copy of the block header forming the longest chain in the network and check the path proof showing which block's Merkle tree their task is linked to. This allows them to cryptographically confirm even from a mobile device that the entire network has approved the validity of that computation.

11. Combining and Splitting Value and Computational Power

Transactions adopt a structure that can have multiple inputs and outputs. This makes it possible to combine or split value and computational power requests into any size. By splitting a large-scale computational task into thousands of independent subtasks and sending them in parallel to different edge clusters, the surplus resources of the entire network can function as a virtual single massive computer.

12. Privacy

Nexus Network solves the privacy issue by separating the disclosure of information into two layers. First, on the blockchain public ledger, it adopts public-key cryptography to completely separate the user's identity information from the public key. Second, to protect the privacy of the computation data itself, it requires a Trusted Execution Environment TEE as the minimum hardware requirement for nodes. As a medium to long-term roadmap, it is designed with the premise of integrating Fully Homomorphic Encryption FHE, building an environment where consumers can have nodes process input data while it remains encrypted, and only they themselves can decrypt the result. 12.5. Threat Model and Attack Vectors This protocol constructs multi-layered cryptographic and cryptoeconomic defenses against major attack vectors threatening network integrity. First, against Lazy Evaluation, where nodes omit computation to save electricity and submit false results, watchers generate Fraud Proofs using state-of-the-art zkVMs RISC Zero / SP1 during the Optimistic Verification challenge period, exposing the disguise 100 percent and fixing the expected value of the attack strictly to negative through the total confiscation of stakes. Furthermore, against Sybil Attacks and Double Spending of Resources, where a single machine disguises itself as multiple nodes and assigns computational resources to multiple tasks simultaneously, the mechanism cryptographically binds the staking required for task acceptance with the physical hardware profile of the node, suppressing the attack by confiscating all stakes the moment timeouts or fraudulent traces associated with double-spending are detected. Finally, regarding the forgery of electricity meters through software tampering, while acknowledging it as an unsolved problem in the current IoT domain, the initial phase physically isolates this attack vector and firmly maintains a trustless execution environment by mandating TEEs and only allowing certified smart meters signed at the hardware level as oracles.

12.1 Hardware Fingerprinting via Micro-benchmarks

To prevent nodes from spoofing high-end GPUs (e.g., reporting a 4090 while using a lower-tier card), Nexus employs Probabilistic Hardware Fingerprinting. During the initial NCU benchmarking, the protocol issues non-deterministic computational micro-tasks unique to specific GPU architectures. These tasks exploit hardware-specific execution quirks and memory-access patterns, creating a digital "fingerprint" that is nearly impossible to replicate via software emulation.

13. Mathematical Foundation and Attack Resistance

The probability that an attacker catches up from a delay of blocks is approximated by the Gambler's Ruin problem and defined as follows. Let be the probability that an honest node finds the next block, and be the probability that an attacker finds the next block.

$$q_z = \begin{cases} 1 & \text{if } p \leq q \\ (q/p)^z & \text{if } p > q \end{cases}$$

As long as the resources of honest nodes exceed those of attackers, the probability of an attack succeeding converges to zero exponentially as the blocks get deeper. Furthermore, with the implementation of slashing that immediately confiscates the staking assets of nodes that submitted invalid proofs, the economic expected value of an attack is always negative.

14. Development Roadmap

The Nexus Network development trajectory is designed as a modular evolution from a functional compute bridge to a global, sovereign energy-backed resource grid. The initial three-month focus (Phase 0: The Nexus Bridge) prioritizes a functional CLI for immediate GPU resource sharing and a basic AI inference relay, establishing a "working block" of utility before constructing the complete architectural castle. This foundation matures into the formal Compute Market (Phase 1: Months 1-6), which standardizes AI inference workloads and launches the high-performance edge node onboarding program to scale network capacity. The subsequent stage (Phase 2: Months 6-12) introduces Verifiable Integrity by fully integrating RISC Zero and SP1 zkVMs for optimistic fraud proofs, while simultaneously enhancing edge clustering logic to minimize latency for real-time applications. The roadmap culminates in the realization of The Energy Nexus (Phase 3: Year 1+), introducing "energy-aware" staking rewards and pilot programs for direct utility grid API integration, finally manifesting the protocol's core vision: a fully autonomous and thermodynamic energy-compute resource pool.

14.1 Join the Infrastructure

Nexus Network is built by people, not just machines. In our early phase, we seek partners who can provide "High-Performance Edge Nodes." If you have a professional workstation or idle GPU power (RTX 3080 series or higher) and a passion for decentralized AI, you are the foundation we are looking for.

Core Technical Stack:

- Language: Rust (for memory safety and high-performance execution).
- P2P Layer: libp2p (for resilient, decentralized node communication).
- Verification: Optimistic Fraud Proofs utilizing RISC Zero or SP1 zkVM for dispute resolution.
- Security: Mandatory TEE (Intel SGX / AMD SEV) for initial supply nodes.

14.2 Phase 0 MVP: The LLM Wedge

Our immediate goal for Phase 0 is not a general-purpose compute market, but a specialized LLM Inference Grid. We will launch with a single, optimized model: Llama-3-8B. By narrowing the scope to one model and a specific Rust-based client, we will achieve a functional marketplace within 90 days, providing AI developers with a sub-API alternative that is 30% cheaper than OpenAI's current pricing.

15. Global Governance and Swiss Institutional Oversight

The governance of the protocol will be attributed to a foundation in the Canton of Zug, Switzerland. To ensure robust legal compliance and institutional-grade oversight under Swiss law, the foundation's board of directors will be composed of experienced professionals in legal, financial, and blockchain governance. While I, the founder, lead the long-term technical and strategic vision as the Founder-Architect, this professional board structure guarantees that the Nexus Network operates with the highest standards of transparency and regulatory accountability.

16. Fairness and Resource Allocation

To ensure the Nexus Network remains a common resource for humanity, the $NXS token distribution is hard-coded for fairness. The founding team will only ever hold 20% of the total supply, subject to a strict 4-year vesting schedule to prove our long-term commitment. The remaining 80% is dedicated entirely to the community: 50% for those who provide computational power (Mining) and 30% for ecosystem growth and developer grants. We win only when the network wins.

16.1 Resource Allocation and Emission Schedule

Category	Allocation	Vesting / Details
Resource Mining (Nodes)	50%	Emission follows a 4-year halving cycle.
Ecosystem & Grants	20%	For builders and open-source AI researchers.
Foundation & Treasury	15%	Multi-sig controlled for long-term governance.
Core Contributors	15%	1-year cliff, 4-year linear vesting.

Max Supply: Fixed at 210,000,000 NXS.

Deflationary Mechanism: 15% of all NCU transaction fees are permanently burned to maintain scarcity and drive value back to the ecosystem.

Nexus Network Governance Mechanism

Depending on future DAO governance decisions, a portion of network fees may be allocated toward ecosystem buyback mechanisms or secondary resource incentives. This ensures the protocol remains adaptive to global regulatory landscapes while maintaining long-term token utility.

17. Conclusion

I have proposed a peer-to-peer resource distribution system that integrates computational power and energy. By using mathematical standardization NCU and limited cryptographic proofs ZKP, it establishes a free and robust resource market excluding central administrators with a realistic load. Nexus Network will democratize the power grid and computation network of the AI era, becoming the foundation for liberating humanity from centralized constraints.

18. Disclaimers

This paper describes a technical concept and does not intend any solicitation of investment. Using this protocol, participating in development, and acquiring tokens involve the following risks. Technical risks: Losses due to bugs or unknown vulnerabilities in smart contracts. Regulatory risks: Service suspensions or specification changes due to changes in legal regulations of various countries. Market risks: Fluctuations in the asset value of tokens accompanying fluctuations in the demand for computational resources. The founder and

development team are not responsible for any direct or indirect damages resulting from this project. Participants shall make judgments at their own risk.

18.5 Glossary of Terms

- NCU (Nexus Computing Unit): The standard metric used to normalize disparate hardware performance and energy efficiency into a single tradable unit.
- ZKP (Zero-Knowledge Proof): A cryptographic method allowing one party to prove a statement is true without revealing the underlying data.
- TEE (Trusted Execution Environment): A secure area of a main processor that ensures data and code are protected with respect to confidentiality and integrity.
- Slashing: A protocol mechanism where a node’s staked tokens are confiscated as a penalty for submitting fraudulent computation results.
- libp2p: A modular network stack used for peer-to-peer communication, ensuring the protocol's censorship resistance.

19. References

[1] Nakamoto, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System*.

[2] Ben-Sasson, E., Chiesa, A., et al. (2014). *SNARKs for C: Verifying Program Executions Succinctly and in Zero Knowledge*. (CRYPTO 2013).

[3] Wood, G. (2014). *Ethereum: A Secure Decentralised Generalised Transaction Ledger.*

[4] Financial Services Agency, Japan. *Payment Services Act (Act No. 59 of 2009)*.

[5] Swiss Federal Council. (2021). *Federal Act on the Adaptation of Federal Law to Developments in Distributed Ledger Technology (DLT Act)*.

[6] Gentry, C. (2009). *Fully Homomorphic Encryption Using Ideal Lattices*. (STOC 2009).

[7] Eberhardt, J., & Tai, S. (2017). *On or Off the Blockchain? Insights on Off-Chaining Computation and Data*.

[8] Androulaki, E., et al. (2018). *Hyperledger Fabric: A Distributed Operating System for Permissioned Blockchains*.

[9] Protocol Labs. (2017). *Filecoin: A Decentralized Storage Network*.

[10] Ben-Sasson, E., et al. (2018). *Scalable, transparent, and post-quantum secure computational integrity*. (zk-STARKs).

[11] Mihaylov, M., et al. (2014). *NRGcoin: Virtual currency for trading of renewable energy in smart grids*.

[12] RISC Zero Team. (2023). *RISC Zero: A Zero-Knowledge Virtual Machine for General Purpose Computing*.

[13] Rao, Y. (2021). *Bittensor: A Peer-to-Peer Intelligence Market*.

[14] Protocol Labs. (2019). *libp2p: A Modular Network Stack for Peer-to-Peer Applications*.

[15] Bennett, C. H. (1982). *The Thermodynamics of Computation – Individual and Society.*

[16] Landauer, R. (1961). *Irreversibility and Heat Generation in the Computing Process*. (IBM Journal of Research and Development).

[17] Buterin, V. (2021). *An Incomplete Guide to Rollups (Optimistic vs. ZK-Rollups)*.

[18] Succinct Labs. (2024). *SP1: A Next-Generation zkVM for High-Performance Proof Generation*.

[19] FINMA. (2018). *Guidelines for enquiries regarding the regulatory framework for initial coin offerings (ICOs)*. (Switzerland).

[20] Landauer, R. (1991). *Information is Physical*. (Physics Today).

[21] Borchers, F., et al. (2023). *Decentralized Training and Inference of Large Language Models: A Survey.*

[22] Landauer, R. (1996). *Physical Limits of Computation*. (Science).

[23] AI@Meta. (2024). *Llama 3 Model Card*. (Meta AI Research).

[24] Borzunov, A., et al. (2022). *Petals: Collaborative Inference and Fine-tuning of Large Models*. (arXiv:2209.01188).

[25] Batori, I., et al. (2022). *GPU Fingerprinting and Hardware-Level Security: A Comprehensive Review*.

[26] Buterin, V., & Griffith, V. (2017). *Casper the Friendly Finality Gadget*. (Ethereum Foundation - *Cost of Corruption and Economic Finality*).

20. The Nexus Manifesto & Builder Invitation

I have laid out the blueprint for the Nexus Network as a definitive architectural document. In this new paradigm, compute is no longer priced by hardware it is priced by energy. While my role as a fifteen-year-old Founder-Architect is to safeguard the long-term vision and thermodynamic logic of this protocol, the transition from theory to a sovereign infrastructure requires "Builders"—highly skilled engineers and cryptographers who possess the technical mastery to translate these formulas into a resilient, scalable codebase.

We are seeking core contributors who are not looking for a traditional job, but for a mission to dismantle the current centralized monopoly of AI and energy. Ideal candidates must demonstrate deep proficiency in low-level systems (Rust/C++), decentralized networking (libp2p), or the design of ZK-circuits using frameworks such as RISC Zero and SP1.

We demand a rigorous alignment with the principles of decentralized trust and a commitment to our four-year vesting schedule, reflecting our focus on a decade-scale horizon rather than short-term market gains. This is a call to those who recognize that the current AI infrastructure is a historical anomaly that must be corrected.

If you possess the expertise to challenge or improve this architecture, I invite you to submit a Technical Critique of this paper—identifying inefficiencies or proposing superior mechanisms—or a Proof of Work (PoW) to yizhenxianshi@gmail.com with the subject line [Core Builder Application].

We do not want your résumé; we want your insight. Help us build the first block of the global resource grid.

21. Legal Framework and Disclaimer

Nexus Network is currently in its conceptual and architectural phase. The proposed legal and governance structures are intended to be established as the protocol transitions to its implementation phase.

Nexus Network operates under the jurisdiction of the Canton of Zug, Switzerland, leveraging the Swiss DLT Act to ensure a robust and transparent legal foundation for decentralized governance. The project is managed by the Nexus Network Foundation, which is committed to regulatory compliance with the Swiss Financial Market Supervisory Authority (FINMA) as a Virtual Asset Service Provider (VASP). The Nexus token (NXS) is strictly classified as a utility token, serving as the essential unit for the normalization and transaction of computational resources within the ecosystem.

This document is provided solely for technical information and does not constitute an offer to sell or a solicitation of an offer to purchase any securities or financial instruments in any jurisdiction. Engaging with the development or use of this protocol involves significant technical and market risks; therefore, the founder and the foundation shall not be held liable for any direct or indirect damages arising from the use of the information contained herein. All participants are advised to exercise their own professional judgment and assume full responsibility for their involvement in this evolving infrastructure.

22. Final Provisions: The Architect's Mandate, Swiss Governance, and Legal Framework

The architecture presented in this document serves not merely as a technical proposal but as a definitive blueprint for the next generation of human infrastructure, designed to resolve the fundamental contradiction between the centralization of AI intelligence and the physical reality of energy scarcity. As the Founder-Architect, my objective is to safeguard this long-term vision; however, transforming this blueprint into a sovereign, living protocol requires the hands of "Builders"—the 0.1% of engineers, cryptographers, and researchers who view the current monopoly of big-tech as a historical anomaly that must be corrected. We are not offering traditional employment, but a position as a Core Contributor demanding technical excellence in low-level systems proficiency (Rust/C++) for the NCU Normalization Engine, mastery of decentralized P2P networking (libp2p), applied cryptography for designing zkVM circuits (RISC Zero, SP1), and a deep understanding of thermodynamic computing. If you possess the expertise to challenge this architecture, do not send a résumé; instead, submit a Technical Critique identifying inefficiencies or proposing superior mechanisms to yizhenxianshi@gmail.com with the subject line `[Core Builder Application] - Technical Critique`.

To ensure the long-term survival and institutional-grade oversight of this protocol, the Nexus Network will be governed by the Nexus Network Foundation, a non-profit entity to be incorporated in the Canton of Zug, Switzerland, leveraging Switzerland's world-leading Swiss DLT Act to provide an ideal regulatory environment. Given my status as a fifteen-year-old visionary, I will serve strictly as the Founder-Architect, focusing exclusively on the technical roadmap and strategic vision, while the Foundation's Board of Directors will be composed of seasoned professionals with backgrounds in international law, Swiss finance, and high-scale blockchain governance. This dual structure guarantees that the protocol remains innovative yet legally beyond reproach as it transitions over a five-year horizon toward a fully Decentralized Autonomous Organization(DAO).

Within this governance framework, it must be explicitly understood that the Nexus Network is currently in its Conceptual and Architectural Phase, and all descriptions herein are subject to change based on technical feasibility and regulatory requirements. The $NXS token is strictly classified as a Utility Token under Swiss FINMA guidelines, serving solely as the essential medium for the normalization and transaction of computational power and energy, and is not an investment contract, security, or financial instrument. Participants must acknowledge the inherent risks of experimental technologies, including potential smart contract vulnerabilities, sudden regulatory shifts, and oracle failures related to the physical tampering of IoT devices. Furthermore, due to the Founder's minority status (15 years old), all legal contracts and formal institutional agreements will be executed by legally appointed representatives and professional board members in accordance with Swiss law. TO THE MAXIMUM EXTENT PERMITTED BY APPLICABLE LAW, THE FOUNDER, THE FOUNDATION, AND ALL CONTRIBUTORS SHALL NOT BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, OR CONSEQUENTIAL DAMAGES ARISING FROM THE USE OF THIS PROTOCOL OR RELIANCE ON THE INFORMATION CONTAINED IN THIS DOCUMENT; ALL SOFTWARE AND CONCEPTS ARE PROVIDED "AS-IS" WITHOUT WARRANTY OF ANY KIND.

22.1 Operational Transparency and Foundational Integrity

At the time of this architectural disclosure, the Nexus Network is in its conceptual inception phase. As the Founder-Architect is 15 years old, he does not currently possess the unilateral legal capacity required to execute institutional contracts or manage corporate assets independently under Swiss jurisdiction. To ensure the highest standards of accountability, all formal legal actions and financial management will be handled by a professional Board of Directors upon the official incorporation of the Nexus Network Foundation in Zug, Switzerland. The term "We" throughout this document represents the future collective of core builders and the decentralized community that this manifesto aims to mobilize.

22.2 The Path to Autonomy

The Nexus Network is designed as a sovereign infrastructure that will eventually transition over a five-year horizon toward a fully Decentralized Autonomous Organization (DAO). This transition ensures that the protocol remains a neutral, common resource for humanity, governed by its users rather than any single individual or corporation.

Copyright Notice:

This document is the intellectual property of the Nexus Network Project. While the Nexus Network Foundation is being formally established in the Canton of Zug, Switzerland, all original concepts, NCU formulas, and architectural designs contained herein are protected under international copyright law from the date of initial publication (March 28, 2026).

(C) 2026 Nexus Network Foundation. All Rights Reserved.